

MAP THE WHOLE UNIT ON ONE SCREEN — 22 SYLLABUS HEADS			
Foundations 01–04 Importance · classification · tabulation · parameter/statistic	Presentation 05 Diagrams & graphs; histogram, ogive, pie	Central tendency 06–07 Mean, median, mode (simple + grouped)	Dispersion 08–09 Range, QD, MD, variance, SD, CV, SE
Shape 10 Moments, skewness, kurtosis	Probability 11 Theorems, addition, multiplication	Distributions 12–13 Binomial, Poisson, Normal, SNV	Relationship 14–15 Correlation (r , r_s), regression lines
Sampling 16 SRS, stratified, systematic, cluster	Large sample 17 Z-test (mean, 2 means, proportion)	Small sample 18 t-test: one, two, paired	Attributes 19 χ^2 goodness of fit & independence
Designs 20 3 principles · CRD · RBD	ANOVA 21 Partition of SS · F-test · CD	Non-parametric 22 Sign, Mann-Whitney, Kruskal-Wallis	Computer 23–25 Languages · DBMS · MS-Office · Excel · net

★ HIGHEST-YIELD IN THIS UNIT

- ▶ **χ^2 test** — asked every year (genetics ratios). Formula + df + Yates.
- ▶ **CRD vs RBD** — the standard 10-mark differentiate.
- ▶ **ANOVA table** — write the skeleton with df column correctly.
- ▶ **Normal distribution** — properties list + area rule.
- ▶ **Correlation vs Regression** — table + $r = \sqrt{(b_{yx} \cdot b_{xy})}$.
- ▶ **CV** — why unit-free; used to compare two herds.
- ▶ **SD vs SE** — the single most confused pair.
- ▶ **Binomial vs Poisson vs Normal** — mean/variance relations.

1 WHY BIostatISTICS · SCOPE & LIMITATIONS

Statistics = science of collection, organisation, presentation, analysis and interpretation of numerical data. **Biostatistics** = statistics applied to biological, medical and veterinary data (Bio + Statistics; coined from the variability of living material).

- ▶ **Why needed:** biological data are inherently **variable** — no two animals give identical yield; conclusions must be drawn from samples, not whole populations.
- ▶ **Uses in vet science:** comparing ratios/treatments · estimating heritability & breeding values · designing field trials · disease prevalence & morbidity-mortality rates · standardising lactation yield · drug efficacy trials · sire evaluation.
- ▶ **Limitations:** deals with **aggregates, not individuals**; only quantitative facts; results are true only **on average**; can be misused; requires expert handling.

DEFINITION TO QUOTE

Croxtan & Cowden: "Statistics is the science of collection, presentation, analysis and interpretation of numerical data."

2 POPULATION VS SAMPLE — THE 3 WORDS THEY ASK

Term	Meaning	Symbol
Observation	A single recorded value / measurement on one unit (one cow's milk yield)	x_1
Parameter	A constant describing the whole population ; usually unknown, fixed	$\mu, \sigma, \sigma^2, P, \rho$
Statistic	A value computed from the sample ; varies sample to sample; estimates the parameter	$\bar{x}, s, s^2, p, r$

TRAP

Greek = population, Roman = sample. A statistic is a **variable** (it changes with the sample); a parameter is a **constant**. Sampling distribution of a statistic → gives the **standard error**.

Variable types: **Qualitative / attribute** breed, sex, coat colour

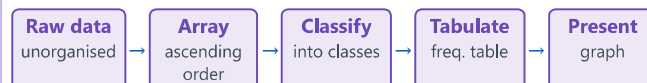
Quantitative → **Discrete** (litter size, counts) & **Continuous** (milk yield, body weight).

Scales: Nominal → Ordinal → Interval → Ratio.

3 CLASSIFICATION & TABULATION OF DATA

Classification = arranging raw data into homogeneous groups.

Tabulation = systematic presentation of classified data in rows and columns.



Basis of classification	Example
Geographical (spatial)	Milk production state-wise
Chronological (temporal)	Yearly egg production
Qualitative (attribute)	Breed, sex, coat colour
Quantitative (variable)	Body weight classes

Frequency-table terms: class limits (lower/upper) · class boundaries (true limits, ± 0.5 in exclusive → inclusive conversion) · **class interval i = $(L-S)/\text{no. of classes}$** · mid-point/class mark = $(LL+UL)/2$ · **Sturges' rule $k = 1 + 3.322 \log_{10} N$** · cumulative frequency (less-than / more-than) · relative frequency.

Parts of a table: table number, title, head-note, stub (rows), caption (columns), body, footnote, source note.

4 DIAGRAMMATIC & GRAPHICAL REPRESENTATION

Device	Used for / key point
Bar diagram (simple, multiple, sub-divided/component, percentage)	Discrete / qualitative data. Bars are separated , width uniform, only length matters.
Pie / circle diagram	Component parts of a whole. Angle = $(\text{component}/\text{total}) \times 360^\circ$.
Pictogram · Cartogram	Popular presentation; map-based (cartogram).
Histogram	Continuous grouped data. Bars adjacent (no gap); area $\propto$ frequency . Mode located graphically from it.
Frequency polygon	Mid-points joined by straight lines; area = histogram area.
Frequency curve	Smoothed polygon; large N.
Ogive (cumulative freq. curve)	Less-than & more-than ogives; their intersection = MEDIAN . Also gives quartiles & percentiles.
Scatter diagram	First step in correlation; shows direction & linearity.

ONE-MARK TRAP

Histogram = **continuous, no gaps, area $\propto$ f**. Bar diagram = **discrete, gaps present, length $\propto$ f**. Median from **ogive**; Mode from **histogram**.

5 MEASURES OF CENTRAL TENDENCY — FORMULA VAULT

A single value around which the data cluster. **Requisites of an ideal average (Yule):** rigidly defined · based on all observations · easy to understand & compute · suitable for further algebraic treatment · least affected by sampling fluctuation · not unduly affected by extremes.

MEAN simple $\bar{x} = \Sigma x / n$

MEAN grouped $\bar{x} = \Sigma fx / \Sigma f$ (x = class mid-point)

Step-deviation $\bar{x} = A + (\Sigma fd / \Sigma f) \times c$, $d = (x - A)/c$

MEDIAN simple odd: $[(n+1)/2]^{\text{th}}$ item · even: mean of $(n/2)$ & $(n/2 + 1)^{\text{th}}$

MEDIAN grouped $M = L + [(N/2 - cf) / f] \times h$

MODE grouped $Z = L + [(f_1 - f_0) / (2f_1 - f_0 - f_2)] \times h$

Empirical relation **Mode** = 3 **Median** - 2 **Mean**

Quartiles $Q_1 = L + [(N/4 - cf)/f]h$; Q_3 uses $3N/4$

Other averages: **GM** = $\sqrt[n]{x_1 x_2 \dots x_n}$ = antilog($\Sigma \log x / n$) — used for **rates of growth, ratios**. **HM** = $n / \Sigma (1/x)$ — used for **average speed / rates per unit**. Relation: **AM** ≥ **GM** ≥ **HM**; for two values $GM^2 = AM \times HM$.

6 WHICH AVERAGE? — DECISION TABLE

	Mean	Median	Mode
Based on all obs.	Yes	No (positional)	No (positional)
Affected by extremes	Yes, badly	No	No
Algebraic treatment	Yes	No	No
Open-end classes	Cannot compute	Can	Can
Qualitative data	No	Ordinal only	Yes
Graphic location	—	Ogive	Histogram
Value may not exist	Yes (2.4 calves)	Actual value	Actual value
Uniqueness	Unique	Unique	May be bi/multi-modal or absent

PROPERTIES OF THE MEAN — DIRECT 2-MARK

1. $\Sigma(x - \bar{x}) = 0$ (algebraic sum of deviations from mean is zero). 2. $\Sigma(x - \bar{x})^2$ is **minimum** (least-squares property). 3. Combined mean $\bar{x}_{12} = (n_1 \bar{x}_1 + n_2 \bar{x}_2) / (n_1 + n_2)$. 4. Affected by change of both origin and scale.

TRAP

In a **positively skewed curve**: **Mode < Median < Mean**. Negatively skewed: **Mean < Median < Mode**. Symmetrical: all three coincide.

7 MEASURES OF DISPERSION — FORMULA VAULT

Dispersion / variation = extent to which observations scatter about the average. Two herds may have the same mean but very different uniformity — dispersion measures that.

Measure	Absolute	Relative (coefficient)
Range	$R = L - S$	$(L - S) / (L + S)$
Quartile deviation (semi-interquartile)	$QD = (Q_3 - Q_1) / 2$	$(Q_3 - Q_1) / (Q_3 + Q_1)$
Mean deviation	$MD = \Sigma x - A / n$	MD / A
Standard deviation	$s = \sqrt{[\Sigma(x - \bar{x})^2 / (n-1)]}$	$CV = (s / \bar{x}) \times 100$

Variance (short-cut) $s^2 = [\Sigma x^2 - (\Sigma x)^2 / n] / (n - 1)$

Grouped variance $s^2 = [\Sigma fx^2 - (\Sigma fx)^2 / \Sigma f] / (\Sigma f - 1)$

Standard error of mean $SE(\bar{x}) = s / \sqrt{n}$

Normal-curve relation $QD : MD : SD = 0.6745 : 0.7979 : 1 \approx 10 : 12 : 15$

Why (n - 1)? Divisor **n - 1 = degrees of freedom**; it makes s^2 an **unbiased estimator** of σ^2 . Population variance uses N .

SD properties: independent of **change of origin** but **not of scale**; minimum when deviations taken from mean; SD of combined series can be pooled.

8 CV · SD VS SE — THE EXAMINER'S FAVOURITE PAIR

	Standard Deviation (SD)	Standard Error (SE)
Measures	Variability of individual observations about the sample mean	Variability of the sample mean about the population mean
Formula	$\sqrt{[\Sigma(x - \bar{x})^2 / (n-1)]}$	$s / \sqrt{n}$
Effect of 1 n	Stays ~constant	Decreases ($\propto 1/\sqrt{n}$)
Used for	Describing spread of data	Confidence limits & tests of significance

COEFFICIENT OF VARIATION — FULL ANSWER

CV = (SD/Mean) × 100. It is a **relative, unit-free (percentage)** measure, so it is the **only valid way to compare variability of two series with different units or widely different means** (e.g. milk yield in kg vs fat %; Sahiwal vs HF herd). **Lower CV = more consistent / uniform / stable**. Used in animal breeding to judge uniformity of a herd, repeatability of a lab technique, and in field trials CV of the experiment indicates **experimental precision** (a low CV in ANOVA = well-conducted trial).

TRAP

CV is meaningless if the mean is near zero or negative. Range is the **simplest but most unstable** (uses only 2 values). SD is the **best and most widely used** measure of dispersion.

9 MOMENTS · SKEWNESS · KURTOSIS — SHAPE OF THE DISTRIBUTION

Moments describe the shape of a frequency distribution — arithmetic means of powers of deviations.

Central moments $\mu_r = \Sigma f(x - \bar{x})^r / N$

$\mu_1 = 0$ always · $\mu_2 = \text{variance } (\sigma^2)$ · $\mu_3 \rightarrow \text{skewness}$ · $\mu_4 \rightarrow \text{kurtosis}$. Raw moments μ'_r are taken about an arbitrary point A ; convert with $\mu_2 = \mu'_2 - \mu'_1{}^2$, $\mu_3 = \mu'_3 - 3\mu'_2\mu'_1 + 2\mu'_1{}^3$.

β_1 (skewness) $\beta_1 = \mu_3^2 / \mu_2^3$; $\gamma_1 = \sqrt{\beta_1} = \mu_3 / \sigma^3$

β_2 (kurtosis) $\beta_2 = \mu_4 / \mu_2^2$; $\gamma_2 = \beta_2 - 3$

SKEWNESS = lack of symmetry (which tail is longer).

Karl Pearson $S_k = (\text{Mean} - \text{Mode}) / SD = 3(\text{Mean} - \text{Median}) / SD$

Bowley (quartile) $S_k = (Q_3 + Q_1 - 2M) / (Q_3 - Q_1)$

Type	Order	Tail	S_k
Symmetrical	Mean = Median = Mode	Both equal	0
Positive skew	Mode < Median < Mean	Longer to right	+ve
Negative skew	Mean < Median < Mode	Longer to left	-ve

KURTOSIS = peakedness / flatness of the curve relative to the normal curve.

Curve	β_2	γ_2	Look
Mesokurtic	= 3	0	Normal
Leptokurtic	> 3	+ve	More peaked, thin tails
Platykurtic	< 3	-ve	Flat-topped, broad

MNEMONIC

"LEAP high, PLATE flat." Leptokurtic = leaps up (peaked). Platykurtic = flat like a plate. Normal curve: $\beta_1 = 0$ and $\beta_2 = 3$.

10 PROBABILITY — DEFINITIONS & THEOREMS

Classical $P(A) = \text{favourable outcomes} / \text{total equally-likely outcomes} = m/n$

$0 \leq P(A) \leq 1$ · $P(\text{certain}) = 1$ · $P(\text{impossible}) = 0$ · $P(A) + P(\bar{A}) = 1$ (complement).

Empirical / relative-frequency definition: $P = \lim (m/n)$ as $n \rightarrow \infty$ (von Mises).

Terms: random experiment · trial · sample space (S) · event · **mutually exclusive** (cannot occur together, $A \cap B = \emptyset$) · **exhaustive** (cover S) · **equally likely** · **independent** (one does not affect the other).

ADDITION (OR) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

— mutually exclusive $P(A \cup B) = P(A) + P(B)$

MULTIPLICATION (AND) $P(A \cap B) = P(A) \cdot P(B|A)$

— independent $P(A \cap B) = P(A) \cdot P(B)$

Conditional $P(B|A) = P(A \cap B)/P(A)$

Bayes' theorem $P(A_i|B) = P(A_i)P(B|A_i) / \sum P(A_j)P(B|A_j)$

GENETICS APPLICATION (WRITE THIS)

Monohybrid $Aa \times Aa \rightarrow P(aa) = 1/4$ (classical). Both parents' gametes are independent $\rightarrow$ multiplication rule. Probability that a calf is **either red or polled** $\rightarrow$ addition rule. Mendel's laws are essentially statements of **independent probability**, which is why probability underpins all of animal genetics.

11 BINOMIAL DISTRIBUTION (BERNOULLI, 1713)

pmf $P(x) = {}^nC_x p^x q^{n-x}$, $x = 0, 1, 2, \dots, n$; $q = 1 - p$

Mean $\mu = np$ Variance $\sigma^2 = npq$ SD $= \sqrt{npq}$

- **Conditions:** fixed number of trials n · each trial has only **two mutually exclusive outcomes** (success/failure) · **p constant** for every trial · trials are **independent**.
- **Discrete** distribution; parameters **n and p**.
- **Mean > Variance** always (because $q < 1$). **This is the identifying line.**
- Symmetrical when $p = q = 0.5$; positively skewed when $p < 0.5$.
- Skewness $\gamma_1 = (q - p)/\sqrt{npq}$; Kurtosis $\beta_2 = 3 + (1 - 6pq)/npq$.
- **Vet examples:** sex ratio of calves born · number of animals responding to a vaccine out of n · Mendelian segregation ratios · number of carriers in a litter.

12 POISSON DISTRIBUTION (1837) — LAW OF RARE EVENTS

pmf $P(x) = e^{-\lambda} \lambda^x / x!$, $x = 0, 1, 2, \dots, \infty$

Key identity Mean = Variance = $\lambda = np$ (SD = $\sqrt{\lambda}$)

- **Limiting case of binomial** when $n \rightarrow \text{very large}$, $p \rightarrow \text{very small}$, $np = \lambda$ finite.
- Single parameter λ ; discrete; range $0 \rightarrow \infty$ (no upper limit, unlike binomial).
- Always **positively skewed**; becomes symmetrical as λ increases; $\gamma_1 = 1/\sqrt{\lambda}$, $\beta_2 = 3 + 1/\lambda$.
- Sum of independent Poisson variates is Poisson (additive property).
- **Vet examples:** number of **rare congenital defects** per 1000 births · deaths from a rare disease · bacterial colonies per plate · somatic cell counts · number of ticks per animal · accidents in a farm per month.

ONE-MARK DISCRIMINATOR

Binomial: **mean > variance**, finite range $0-n$, two parameters. Poisson: **mean = variance**, infinite range, one parameter. If a question says "rare event / very large n / small p " $\rightarrow$ **Poisson**.

13 NORMAL DISTRIBUTION (DE MOIVRE · LAPLACE · GAUSS)

pdf $f(x) = 1/(\sigma\sqrt{2\pi}) \cdot e^{-(x-\mu)^2/2\sigma^2}$, $-\infty < x < \infty$

PROPERTIES — write all of these for full marks:

- **Continuous**; parameters μ (**location**) and σ (**scale**); written $X \sim N(\mu, \sigma^2)$.
- **Bell-shaped, perfectly symmetrical** about $x = \mu$; **unimodal**.
- **Mean = Median = Mode = μ** ; skewness $\beta_1 = 0$, kurtosis $\beta_2 = 3$ (**mesokurtic**).
- **Total area under the curve = 1** (or 100%); curve never touches the X-axis — **asymptotic** to it.
- Quartiles equidistant: $Q_3 - M = M - Q_1$; **QD = 0.6745σ** , **MD = 0.7979σ** .
- Points of **inflection at $\mu \pm \sigma$** . Odd central moments = 0; $\mu_4 = 3\sigma^4$.
- Linear combination of normal variates is normal. **Central Limit Theorem:** means of samples from ANY population tend to normal as n increases.

AREA RULE — MEMORISE EXACTLY

$\mu \pm 1\sigma = 68.27\%$ · $\mu \pm 2\sigma = 95.45\%$ · $\mu \pm 3\sigma = 99.73\%$

Reverse (used in testing): $\mu \pm 1.96\sigma = 95\%$ · $\mu \pm 2.58\sigma = 99\%$ · $\mu \pm 1.645\sigma = 90\%$.

14 STANDARD NORMAL VARIATE & CURVE

SNV $Z = (x - \mu) / \sigma \rightarrow Z \sim N(0, 1)$

- Standardisation converts **any** normal curve to **one** standard curve with **mean 0 and SD 1**, so a single table serves all problems.
- Z is a **pure number** (unit-free) — it tells how many SDs an observation lies from the mean.
- Area is symmetrical: $P(Z > 1.96) = 0.025$; $P(-1.96 < Z < 1.96) = 0.95$.
- **Uses:** converting raw scores · finding % of animals above/below a cut-off (basis of **selection intensity** in breeding!) · normal approximation to binomial (when $np > 5$ and $nq > 5$) and to Poisson ($\lambda > 10$) · basis of all large-sample tests.

LINK TO UNIT 3

The **selection differential** and **intensity of selection (i)** come straight from the normal curve: if the top $p\%$ are selected, **$i = z/p$** , where z is the height of the ordinate at the truncation point. So Unit 1's normal curve is Unit 3's selection theory.

15 THE THREE DISTRIBUTIONS — SIDE BY SIDE

	Binomial	Poisson	Normal
Nature	Discrete	Discrete	Continuous
Parameters	n, p	λ	μ, σ
Range	0 to n	0 to ∞	$-\infty$ to $+\infty$
Mean	np	λ	μ
Variance	npq	λ	σ^2
Mean vs Var	Mean > Var	Mean = Var	Independent
Skewness	$(q-p)/\sqrt{npq}$	$1/\sqrt{\lambda}$ (always +ve)	0
Typical use	Sex ratio, Mendelian ratio	Rare defects, colony counts	Body wt, milk yield, all metric traits

CHAIN OF REASONING

Binomial — (n large, p small) $\rightarrow$ **Poisson** and **Binomial** — (n large, $p \approx 0.5$) $\rightarrow$ **Normal**; **Poisson** — (λ large) $\rightarrow$ **Normal**. Normal is the limiting form of both.

16 CORRELATION

Correlation = degree and direction of **linear association** between two variables. Studied by scatter diagram (visual), Karl Pearson's r (quantitative), and Spearman's rank r_s .

$$\text{Karl Pearson } r = \frac{\sum(x-\bar{x})(y-\bar{y})}{\sqrt{[\sum(x-\bar{x})^2 \cdot \sum(y-\bar{y})^2]}}$$

$$\text{Working form } r = \frac{\text{Cov}(x,y)}{(\sigma_x \cdot \sigma_y)} = \frac{[n\sum xy - \sum x \sum y]}{\sqrt{[n\sum x^2 - (\sum x)^2][n\sum y^2 - (\sum y)^2]}}$$

Spearman rank $r_s = 1 - \frac{6\sum d^2}{n(n^2 - 1)}$, d = difference of ranks

$$\text{SE / test } t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}, \text{ df} = n - 2$$

- **Limits:** $-1 \leq r \leq +1$. $+1$ = perfect positive, -1 = perfect negative, 0 = no linear relation.
- r is a **pure number**, independent of **units** and of **change of origin and scale**.
- r^2 = **coefficient of determination** = proportion of variation in Y explained by X ($r = 0.8 \rightarrow 64\%$ explained).
- **Types:** positive/negative • linear/non-linear • simple, partial, multiple.
- **Spearman** is used for **ranked/ordinal or non-normal** data (e.g. judging scores of animals in a show) — it is the **non-parametric** counterpart.

TRAP

Correlation $\neq$ causation. A high r may be due to a **spurious/nonsense correlation** or a common third cause. Also $r = 0$ only rules out a **linear** relation, not a curvilinear one.

17 REGRESSION

Regression (Sir Francis Galton) = the average functional relationship used to **predict** the dependent variable from the independent one. Lines are fitted by the **method of least squares** (minimising $\sum$ of squared deviations).

$$Y \text{ on } X \quad (Y - \bar{Y}) = b_{yx}(X - \bar{X}) ; b_{yx} = \frac{\sum xy}{\sum x^2} = r \cdot (\sigma_y / \sigma_x)$$

$$X \text{ on } Y \quad (X - \bar{X}) = b_{xy}(Y - \bar{Y}) ; b_{xy} = \frac{\sum xy}{\sum y^2} = r \cdot (\sigma_x / \sigma_y)$$

Bridge to r $r = \pm \sqrt{b_{yx} \cdot b_{xy}} \rightarrow r$ is the GM of the two b 's

- Both regression coefficients carry the **same sign** as r ; both cannot exceed 1 numerically at the same time; their product ≤ 1 .
- The two lines **always intersect at $(\bar{X}, \bar{Y})$** .
- $r = \pm 1 \rightarrow$ the two lines **coincide**; $r = 0 \rightarrow$ they are **perpendicular** (parallel to the axes).
- b is **independent of change of origin but NOT of scale** (opposite of r , which is independent of both).
- b has **units** (kg per day, etc.); r is unit-free.

	Correlation	Regression
Purpose	Measures degree of relation	Establishes nature & predicts
Symmetry	$r_{xy} = r_{yx}$ — one value	$b_{yx} \neq b_{xy}$ — two lines
Units	None (pure number)	Has units
Cause-effect	Does not imply	Implies dependent/independent
Origin & scale	Independent of both	Independent of origin only

18 SAMPLING METHODS — INTRODUCTION

Population (universe) = entire group of interest; **sample** = representative subset; **sampling frame** = complete list of units; **sampling unit**; **sampling fraction** = n/N .

Why sample? saves **time, cost, labour** • the only option when the test is **destructive** (slaughter, necropsy) or the population is infinite • allows greater **depth and accuracy** per unit • gives an estimate of **sampling error**.

Sampling error = difference between sample estimate and population value; **decreases as n increases** and is measurable. **Non-sampling error** = faulty frame, measurement, recording, non-response; **increases with n** and is NOT measurable. **Essentials of good sampling:** representativeness, adequacy, independence, homogeneity.

TRAP

Complete enumeration = **census** (no sampling error, but costly). Only **probability (random) sampling** permits valid statistical inference and computation of standard error.

Method	How it works	When used / note
Simple random (SRS)	Every unit has an equal and independent chance; lottery method or random-number tables; with or without replacement	Population homogeneous ; needs a complete frame
Stratified random	Population divided into homogeneous strata (breed, age, sex), then SRS within each stratum ; proportional or optimum allocation	Population heterogeneous ; gives highest precision ; guarantees representation of every subgroup
Systematic	Choose a random start, then take every kth unit , $k = N/n$	Simple, quick, spread evenly; biased if a periodic/cyclic trend exists in the list
Cluster	Population divided into clusters (villages, herds) ; a few whole clusters selected at random	No frame of individuals available; cheapest for field surveys; lower precision
Multistage	Sampling in successive stages: district $\rightarrow$ village $\rightarrow$ household $\rightarrow$ animal	Large-scale livestock censuses / national surveys
Non-probability	Purposive (judgement), quota, convenience, snowball	Quick, but no valid inference ; sampling error cannot be estimated

REMEMBER THE DIFFERENCE

Stratified: strata are **internally homogeneous**, sampling is done **within every stratum**. **Cluster:** clusters are **internally heterogeneous** (mini-populations), and only **some clusters** are taken — **completely**.

19 TESTS OF HYPOTHESIS — THE UNIVERSAL FRAMEWORK (LEARN ONCE, USE FOR Z, T, χ^2 , F)1. State H_0 & H_1

null = "no difference"

2. Fix α

5% or 1%; 1- or 2-tailed

3. Choose test

by data & n

4. Compute statistic

& degrees of freedom

5. Compare with table

at α & df

6. Decide

reject / fail to reject H_0

- H_0 (**null**): assumption of **no difference / no association**; the hypothesis actually tested. H_1 (**alternative**): what we accept if H_0 is rejected.
- **Level of significance (α):** probability of rejecting a true H_0 ; conventionally **5% ($P < 0.05$)** or **1% Confidence level = $1 - \alpha$** .
- **Degrees of freedom** = number of values free to vary = $n -$ (number of constraints).
- **Critical values (2-tailed, normal): 1.96 at 5%, 2.58 at 1%; one-tailed 1.645 at 5%.**

Decision	H_0 is TRUE	H_0 is FALSE
Reject H_0	TYPE I ERROR (α) producer's risk; "false positive"	Correct decision Power = $1 - \beta$
Fail to reject H_0	Correct decision ($1 - \alpha$)	TYPE II ERROR (β) consumer's risk; "false negative"

TRAP

Reducing α **increases** β . The only way to reduce both is to **increase sample size**. If calculated value $>$ table value $\rightarrow$ **significant, reject H_0** . "Significant" means **unlikely to be due to chance**, not "important".

20 WHICH TEST DO I APPLY? — MASTER DECISION FLOW

Is the data QUANTITATIVE (measurements)?	YES → parametric route (assumes normality)	NO — counts / frequencies / attributes → χ² TEST
Sample size?	n > 30 & σ known → Z-TEST	n < 30 & σ unknown → t-TEST
How many groups?	Two groups → Z or t test	Three or more → ANOVA / F-TEST
Assumptions violated / ranks only?	→ NON-PARAMETRIC (Mann-Whitney, Kruskal-Wallis...)	Comparing two variances → F-TEST

21 LARGE SAMPLE — Z TEST (N > 30)

Single mean $Z = (\bar{x} - \mu) / (\sigma/\sqrt{n})$ Two means $Z = (\bar{x}_1 - \bar{x}_2) / \sqrt{(\sigma_1^2/n_1 + \sigma_2^2/n_2)}$ Single proportion $Z = (p - P) / \sqrt{PQ/n}$ Two proportions $Z = (p_1 - p_2) / \sqrt{\hat{p}\hat{q}(1/n_1 + 1/n_2)}$

Assumptions: large sample, population normal (or CLT applies), σ known, random independent sampling.

Decision: $|Z| > 1.96$ → significant at 5%; $|Z| > 2.58$ → significant at 1%.

95% confidence limits: $\bar{x} \pm 1.96 \text{ SE}$.

22 SMALL SAMPLE — STUDENT'S T TEST (W.S. GOSSET, 1908)

① One-sample $t = (\bar{x} - \mu) / (s/\sqrt{n})$, $df = n - 1$ ② Two independent $t = (\bar{x}_1 - \bar{x}_2) / [S_p \sqrt{(1/n_1 + 1/n_2)}]$, $df = n_1 + n_2 - 2$ — pooled variance $S_p^2 = [\Sigma(x_1 - \bar{x}_1)^2 + \Sigma(x_2 - \bar{x}_2)^2] / (n_1 + n_2 - 2)$ ③ Paired $t = \bar{d} / (s_d/\sqrt{n})$, $df = n - 1$ ($n = \text{pairs}$)

• **Assumptions:** ① sample drawn from a **normal** population ② observations **independent & random** ③ σ **unknown**, estimated by s ④ for two-sample: **homogeneity of variances** (test first with F-test).

• **t-curve:** symmetrical, bell-shaped but **flatter with thicker tails** than normal; **depends only on df**; as $df \rightarrow \infty$, $t \rightarrow Z$ (normal).

• **Paired t** is used when the **same animal is measured twice** (before/after treatment) or animals are matched in pairs — it removes between-animal variation and is therefore **more powerful**.

• **Vet uses:** comparing two rations, two drugs, pre- vs post-treatment PCV, comparing a herd mean with a breed standard.

TRAP

Unpaired data with $n_1 \neq n_2 \rightarrow$ **two-sample t**. Same animals before/after → **paired t**. Using unpaired t on paired data wastes information and loses precision.

23 CHI-SQUARE (χ²) TEST — KARL PEARSON, 1900 ★ EVERY YEARStatistic $\chi^2 = \Sigma (O - E)^2 / E$ Expected freq. (contingency) $E = (\text{Row total} \times \text{Column total}) / \text{Grand total}$ Yates' correction (2×2, $df = 1$) $\chi^2 = \Sigma (|O - E| - 0.5)^2 / E$

Application	Degrees of freedom
Goodness of fit (does observed fit an expected ratio?)	$df = n - 1$ ($n = \text{classes}$)
Test of independence / association ($r \times c$ table)	$df = (r - 1)(c - 1)$
Homogeneity of several populations	$(r - 1)(c - 1)$

Conditions for validity: observations must be **independent and random** · data must be **actual frequencies, not percentages or ratios** · **total N > 50** · **every expected frequency ≥ 5** (if not, **pool** adjacent classes) · $\Sigma O = \Sigma E$.

Nature of curve: continuous, **positively skewed**, ranges 0 to ∞, never negative; shape depends only on df; becomes symmetrical as df increases.

TABLE VALUES TO MEMORISE (5%)

$df\ 1 = 3.84$ · $df\ 2 = 5.99$ · $df\ 3 = 7.81$ · $df\ 4 = 9.49$ · $df\ 5 = 11.07$ | At 1%: $df\ 1 = 6.63$, $df\ 2 = 9.21$, $df\ 3 = 11.34$.

THE GENETICS USE — SAY THIS IN THE ANSWER

χ² is the standard test for **Mendelian segregation ratios**: a monohybrid $F_2\ 3 : 1$ ($df = 1$), a dihybrid $F_2\ 9 : 3 : 3 : 1$ ($df = 3$), a test-cross $1 : 1$. It is also used to test a population for **Hardy-Weinberg equilibrium** (Unit 2) — there $df = \text{number of genotypes} - \text{number of alleles}$ (for 2 alleles, $df = 3 - 2 = 1$), because gene frequency is estimated from the data itself.

24 NON-PARAMETRIC (DISTRIBUTION-FREE) TESTS

Tests that make **no assumption about the form (distribution) of the population** and do not test population parameters. Used for **nominal/ordinal data, small samples, skewed data, or when normality/homogeneity fails**.

Non-parametric test	Parametric equivalent
Sign test · Wilcoxon signed-rank	Paired t-test
Mann-Whitney U (Wilcoxon rank-sum)	Two-sample (unpaired) t-test
Kruskal-Wallis H	One-way ANOVA (CRD)
Friedman test	Two-way ANOVA (RBD)
Spearman's rank correlation	Pearson's correlation
Run test	Test of randomness
Chi-square, Median test, Kolmogorov-Smirnov	—

Advantages: no distributional assumption · usable with **ranks, scores, attributes** · valid for very **small n** · simple, quick, less affected by outliers.

Disadvantages: **less powerful / efficient** when parametric assumptions do hold · **wastes information** (uses ranks not magnitudes) · fewer tests available for complex designs.

25 DESIGN OF EXPERIMENTS — FISHER'S 3 PRINCIPLES

RANDOMISATION

removes bias · validates the error term

REPLICATION

gives an estimate of error · 1 precision

LOCAL CONTROL

blocking · reduces experimental error

Other terms: treatment · experimental unit · **experimental error** (variation between units treated alike) · **precision** $\propto 1/SE$. CRD uses randomisation + replication only; RBD adds local control.

	CRD	RBD
Principles	Randomisation + Replication	All three (+ Local control)
Grouping	None — treatments allotted wholly at random	Units grouped into homogeneous blocks ; each block gets every treatment once
Material	Requires homogeneous material (lab / poultry cages)	Suits heterogeneous material with a known gradient
Replication	May be unequal — most flexible	Must be equal (= no. of blocks)
Error df	Maximum (best for very small experiments)	Less — some df go to blocks
Precision	Lower	Higher (block variation removed from error)
Missing plot	No problem — analyse unequal n	Needs a missing-plot formula
Analysis	One-way ANOVA	Two-way ANOVA

MNEMONIC

"**R-R-L**" — **R**andomisation, **R**eplication, **L**ocal control. CRD = R-R only; RBD = R-R-L. (Latin Square adds local control in **two** directions, rows & columns, and needs $t = r = c$.)

26 ANOVA & F-TEST — WRITE THE TABLE FROM MEMORY

ANOVA (R.A. Fisher) = technique of **splitting the total variation** of a data set into components attributable to **assignable causes (treatments, blocks)** and to **chance (error)**, and testing them by the **variance-ratio (F) test**. Used when **3 or more means** are compared — a t-test on every pair would inflate the Type I error.

$$CF \text{ Correction Factor} = (\text{Grand Total})^2 / N$$

$$TSS \Sigma x^2 - CF \quad SST \Sigma (T_i^2 / r) - CF \quad SSB \Sigma (B_j^2 / t) - CF$$

$$SSE \quad TSS - SST \quad (-SSB \text{ in RBD}) \quad MS = SS / df \quad F = MST / MSE$$

Source	df — CRD	df — RBD	SS	MS	F
Blocks	—	$r - 1$	SSB	MSB	MSB/MSE
Treatments	$t - 1$	$t - 1$	SST	MST	MST/MSE
Error	$n - t$	$(r-1)(t-1)$	SSE	MSE	—
Total	$n - 1$	$rt - 1$	TSS	—	—

$$CD / LSD \quad CD = t_{(\alpha, \text{error df})} \times \sqrt{(2 \times MSE / r)}$$

$$CV \text{ of experiment } CV\% = (\sqrt{MSE} / \text{Grand mean}) \times 100$$

Assumptions of ANOVA: ① effects are **additive** ② errors are **independent** ③ errors are **normally distributed** with mean 0 ④ **homogeneity of variances** (homoscedasticity).

F-test: $F = \text{larger variance} / \text{smaller variance} (S_1^2 / S_2^2)$, **always right-tailed**, $df = (n_1 - 1, n_2 - 1)$; $F \geq 1$; also used to test **equality of two population variances** before a two-sample t-test. **F = t^2 when $df_1 = 1$.**

27 COMPUTER APPLICATION — ONE-PASS SWEEP

Computer languages: **Low level** — **machine language** (binary, the only language the CPU understands) and **assembly** (mnemonics; needs an **ASSEMBLER**). **High level** — English-like, machine-independent: FORTRAN, COBOL, C, C++, Java, Python, R; need a **COMPILER** (whole program at once, faster) or an **INTERPRETER** (line by line, easier to debug). **4GL** — SQL.

DBMS: software to **create, store, retrieve, update and manage** data. Hierarchy: **bit** → **byte** → **field** → **record** → **file** → **database**. **Advantages:** controls **data redundancy**; ensures **consistency, integrity, security, sharing**; backup & recovery. **Models:** hierarchical, network, **relational (RDBMS — tables of rows & columns; primary & foreign key; SQL)**, object-oriented. **Examples:** MS-Access, Oracle, MySQL. Vet use: herd, breeding, AI and disease-surveillance records.

MS-OFFICE: **MS-Word** — word processing, **.docx** · **MS-Excel** — **spreadsheet**: data entry, calculation, charts, statistics, **.xlsx** · **MS-PowerPoint** — presentation slides, **.pptx** · **MS-Access** — **database** (RDBMS), **.accdb**.

28 MS-EXCEL FOR BIOLOGICAL DATA · NETWORKS & E-MAIL

Excel structure: workbook → worksheets → **cell** (intersection of a lettered **column** and numbered **row**, e.g. B7); cell reference may be **relative (B7)**, **absolute (\$B\$7)** or **mixed**; formulas begin with =. Rows 1,048,576 × columns 16,384 (A → XFD).

Excel function / tool	Gives
=AVERAGE() =MEDIAN() =MODE()	Central tendency
=STDEV.S() =VAR.S() =COUNT() =SUM()	Dispersion & totals
=CORREL() =SLOPE() =INTERCEPT()	Correlation & regression
=T.TEST() =Z.TEST() =CHISQ.TEST() =F.TEST()	Tests of significance
Data → Data Analysis (ToolPak)	Descriptive Statistics, ANOVA single factor (CRD), two-factor (RBD), Regression, t-Test, F-Test
Charts · Pivot Table · Sort · Filter	Graphs, summarisation, data handling

Networks: **LAN** (single building) → **MAN** (city) → **WAN** (country/global; the Internet is the largest WAN). **Topologies:** bus, star, ring, mesh, tree. Devices: hub, switch, router, modem, NIC.

Internet: global network of networks; protocol suite **TCP/IP**; **WWW** = the information service on it; **HTTP** (web pages), **FTP** (file transfer), **SMTP** (sending mail), **POP3 / IMAP** (receiving mail). **URL**, IP address, domain name, ISP, browser, search engine.

E-mail: username@domain; fields → **To**, **Cc** (carbon copy — visible), **Bcc** (blind — hidden), **Subject**, **Body**, **Attachment**. Advantages: fast, cheap, records kept, attachments, global.

★ LAST-10-MINUTES: OBJECTIVE AMMUNITION + SUBJECTIVE ATTACK PLANS

Father of statistics — R.A. Fisher (ANOVA, design)

Father of biostatistics — Karl

Pearson (χ^2 , r)

t-test — W.S. Gosset ("Student")

Regression — Sir Francis Galton

$\Sigma(x - \bar{x}) = 0$; $\Sigma(x - \bar{x})^2 = \min$

Best dispersion measure — SD;

most unstable — Range

Unit-free measures — r , CV, Z, β_1 , β_2

Median from ogive; **Mode** from histogram

Mode = 3 Median – 2 Mean

AM ≥ GM ≥ HM

QD : MD : SD = 10 : 12 : 15

Normal: $\beta_1 = 0$, $\beta_2 = 3$

$\mu \pm 1.96\sigma = 95\%$, $\mu \pm 2.58\sigma = 99\%$

Binomial: mean > variance

Poisson: mean = variance = λ

SNV: mean 0, SD 1

$r = \pm \sqrt{(b_{yx} \cdot b_{xy})}$

Lines meet at ($\bar{x}$, $\bar{y}$)

$r = 0$ → lines perpendicular

$\chi^2 \text{ df} = (r-1)(c-1)$

$\chi^2 \text{ 5\%, df 1} = 3.84$

Yates' correction — 2×2 table

Paired t df = $n - 1$

Two-sample t df = $n_1 + n_2 - 2$

RBD error df = $(r-1)(t-1)$

F is always right-tailed; $F = t^2$

($df_1 = 1$)

Type I = α ; Type II = β ; Power = $1 - \beta$

SE = $s / \sqrt{n}$ — falls as n rises

Non-parametric for ANOVA —

Kruskal-Wallis

Compiler — whole program;

Interpreter — line by line

SMTP send, **POP3/IMAP** receive

mail; **Bcc** hidden

Q1 "Define χ^2 test. Describe its applications and limitations." → Definition & formula → conditions for validity → goodness of fit ($df = n-1$) with a **3:1 Mendelian worked example** → test of independence ($df = (r-1)(c-1)$) → Yates' correction → table value 3.84 → limitations (needs $E \geq 5$, actual frequencies, $N > 50$, only tests association not its strength).

Q2 "Differentiate CRD and RBD. Give ANOVA table of RBD." → definition of each → layout diagram → the 8-row comparison table on this page → full RBD ANOVA skeleton with df column → CD formula → conclusion on when to prefer each.

Q3 "Normal distribution and its importance." → pdf to all 7 properties → the area rule → $SNV Z = (x - \mu) / \sigma$ → importance (basis of all parametric tests, CLT, selection intensity, quality control).

Q4 "Correlation vs regression." → definitions → both formulas → comparison table → $r = \pm \sqrt{(b_{yx} \cdot b_{xy})}$ → application in genetics (genetic & phenotypic correlation, regression of offspring on parent = heritability, Unit 2).

Q5 "Measures of dispersion / CV." → why needed → 4 absolute + 4 relative measures with formulas → CV as the only cross-unit comparator → herd-uniformity example → SD vs SE table.

CARRY INTO UNIT 2 & 3

Variance → **components of phenotypic variance**; offspring-on-parent regression → **heritability**; correlation → **genetic & phenotypic correlation**; normal curve → **selection intensity**; χ^2 → **Hardy-Weinberg testing**.